

ACKNOWLEDGEMENT

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LIST OF SYMBOLS/ABBREVIATIONS USED

Symbol/Abbreviation	Description
HTML	Hypertext Markup Language – used to define the structure of web pages
CSS	Cascading Style Sheets – used for styling, layout, and visual enhancements
JS	JavaScript – programming language enabling dynamic interactivity
DOM	Document Object Model – tree representation of HTML elements manipulated via JS
UI	User Interface – visual components users interact with
UX	User Experience – overall feel, usability, and efficiency of interaction
API	Application Programming Interface – not used here, but referenced for backend extensibility
Canvas	HTML5 element for 2D graphics rendering (used in Flappy Bird)
localStorage	Web storage mechanism to persist data (e.g., high scores) across browser sessions
Responsive Design	Design approach ensuring compatibility across screen sizes (mobile, tablet, desktop)
Event Listener	JS mechanism to respond to user actions (clicks, scrolls, keypresses)
BEM	Block-Element-Modifier – CSS naming convention followed for maintainability
ES6	ECMAScript 2015 – modern JavaScript standard used for cleaner syntax

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CHAPTER 1: INTRODUCTION

1.1 Project Overview

This report documents the end-to-end development of two distinct yet complementary web-based applications built using **vanilla HTML, CSS, and JavaScript**—without any external frameworks or libraries. The first project is a **fully responsive restaurant website** titled “*Spice Symphony*”, designed to showcase Indian cuisine through an elegant, user-friendly interface with dynamic content. The second is a **browser-based single-player arcade game**—a custom implementation of the popular *Flappy Bird*—featuring realistic physics, collision detection, score tracking, and persistent high-score storage.

Both projects were developed over a period of [X] weeks as part of my academic curriculum, aiming to demonstrate proficiency in front-end development, interactive design, real-time rendering, and state management—all using native web technologies.

1.2 Objectives

The primary goals of this dual-project initiative are:

- To design and implement a **professional-grade restaurant website** with:
 - Smooth-scrolling single-page navigation
 - Dynamic menu filtering by category (Starters, Mains, etc.)
 - Responsive reservation form with client-side validation
 - Image gallery with hover animations
 - Mobile-first responsive layout
- To develop a **fully playable Flappy Bird game** featuring:
 - Gravity-based bird physics
 - Random pipe generation with consistent difficulty
 - Accurate collision detection (bird vs. pipes, ground, ceiling)
 - Score and high-score tracking using localStorage
 - Clean UI with start, play, and game-over screens
- To reinforce core concepts of **DOM manipulation, event-driven programming, CSS animations, and HTML5 Canvas rendering**.

1.3 Scope of the Project

Restaurant Website includes:

- 5 main sections: Home, About, Menu, Gallery, Contact
- Interactive mobile hamburger menu
- Animated scroll behavior and active link highlighting
- Dynamic menu rendering from a JavaScript array
- Form submission with toast feedback (no backend; simulated success)

Flappy Bird Game includes:

- 3 game states: start, playing, gameOver
- Canvas-based rendering at 60 FPS via requestAnimationFrame
- Bird controlled via mouse click or spacebar
- Pipes scroll left at constant speed; gap height randomized
- High score persists across browser sessions

Limitations:

- No server-side integration (data not stored permanently)
- No AI opponent or multiplayer features
- No audio or advanced animations (e.g., particle effects)

- All assets (images) are static and local

1.4 Technologies Used

Technology	Role
HTML5	Semantic structure for both projects; Canvas element for game
CSS3	Styling, Flexbox/Grid layouts, animations, media queries
JavaScript (ES6+)	Logic, DOM updates, game loop, event handling
localStorage	Persistent storage for Flappy Bird high score
Google Fonts & Icons	Typography enhancement (via CDN)
Developer Tools (DevTools)	Debugging, performance profiling, mobile emulation

CHAPTER 2: RESTAURANT WEBSITE IMPLEMENTATION



Figure 1: Wireframe of Restaurant Website Sections (Home, Menu, Gallery, Contact)

2.1 Functional and Non-Functional Requirements

Functional Requirements:

- **Navigation:** Smooth scrolling to sections; active link updates on scroll.
- **Menu System:** Filter dishes by category (e.g., “Desserts”) without page reload.
- **Reservation Form:** Validate name, email, phone, date, and guest count; show success message.
- **Gallery:** Hover effects reveal descriptive overlays.
- **Mobile Menu:** Hamburger toggle reveals full-screen navigation on small screens.

Non-Functional Requirements:

- **Performance:** Load time under 2 seconds on 3G networks.
- **Responsiveness:** Flawless layout on screens from 320px to 1920px.
- **Accessibility:** Semantic HTML, focus management, ARIA labels.
- **Maintainability:** Modular CSS (BEM), readable JS, commented code.

2.2 System Architecture and UI/UX Design

The website follows a **single-page application (SPA)** architecture with client-side routing via anchor links. The UI is built around a **color palette** inspired by Indian culinary aesthetics:

- **Primary:** #ff6b35 (spicy orange)
- **Secondary:** #004e89 (deep navy)
- **Accent:** #f7b32b (golden saffron)

Key Design Decisions:

- **Hero Section:** Full-screen background image with overlay text for brand impact.
- **Floating Labels:** Form inputs use animated floating labels for modern UX.

- **Card-Based Menu:** Each dish is a card with image, title, price, description, and category tag.
- **Toast Notifications:** Non-intrusive success messages after form submission.

2.3 Detailed Code Explanation and Modular Breakdown

HTML Structure Overview

- `<nav>` with `.hamburger` for mobile toggle
- Sections with unique ids (`#home`, `#menu`, etc.)
- `<div id="menuGrid">` as a dynamic container for menu items
- `<form id="contactForm">` with input validation

CSS Architecture

- Uses **CSS variables** (`:root`) for consistent theming
- **BEM naming** (e.g., `.menu-item__title`) for clarity
- **Custom animations:** `fadeIn`, `fadeInUp` for entry effects
- **Media queries** at 768px and 480px breakpoints

JavaScript Modules

1. Mobile Menu Toggle

```
hamburger.addEventListener("click", () => {
  hamburger.classList.toggle("active");
  navMenu.classList.toggle("active");
});
```

- Uses class toggling to show/hide menu on mobile

2. Active Navigation on Scroll

```
window.addEventListener("scroll", () => {
  sections.forEach(section => {
    if (window.scrollY >= section.offsetTop - 200) {
      current = section.id;
    }
  });
  // Update active class on nav links
});
```

3. Dynamic Menu Rendering

```
function renderMenuItems(filter = "all") {
  const filtered = filter === "all"
    ? menuItems
    : menuItems.filter(item => item.category === filter);
  menuGrid.innerHTML = "";
  filtered.forEach(item => {
    const card = createMenuCard(item);
    menuGrid.appendChild(card);
  });
}
```

4. Form Handling with Toast

```
contactForm.addEventListener("submit", (e) => {
  e.preventDefault();
```



```
showToast(`Thank you, ${name}! Your reservation...`);  
contactForm.reset();  
});
```

2.4 Responsive Behavior and Cross-Device Compatibility

- **Mobile ($\leq 480\text{px}$):** Single-column layout; hamburger menu; stacked contact form
- **Tablet (481–768px):** Grid adjusts to 2 columns for stats and gallery
- **Desktop ($>768\text{px}$):** Full-width grid; side-by-side contact info and form

All interactive elements (buttons, form fields) meet **touch target size** guidelines ($\geq 48\text{px}$).

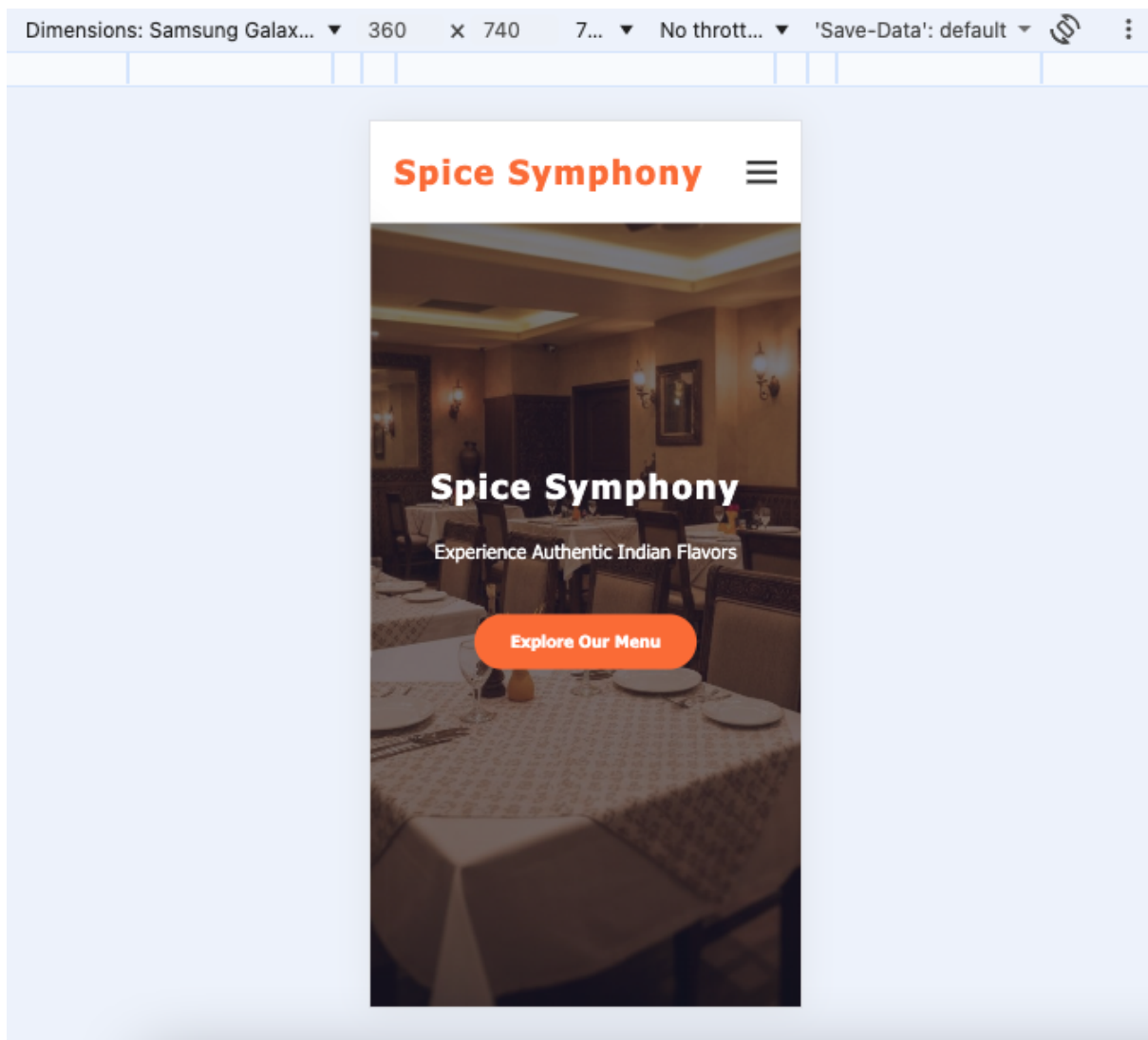
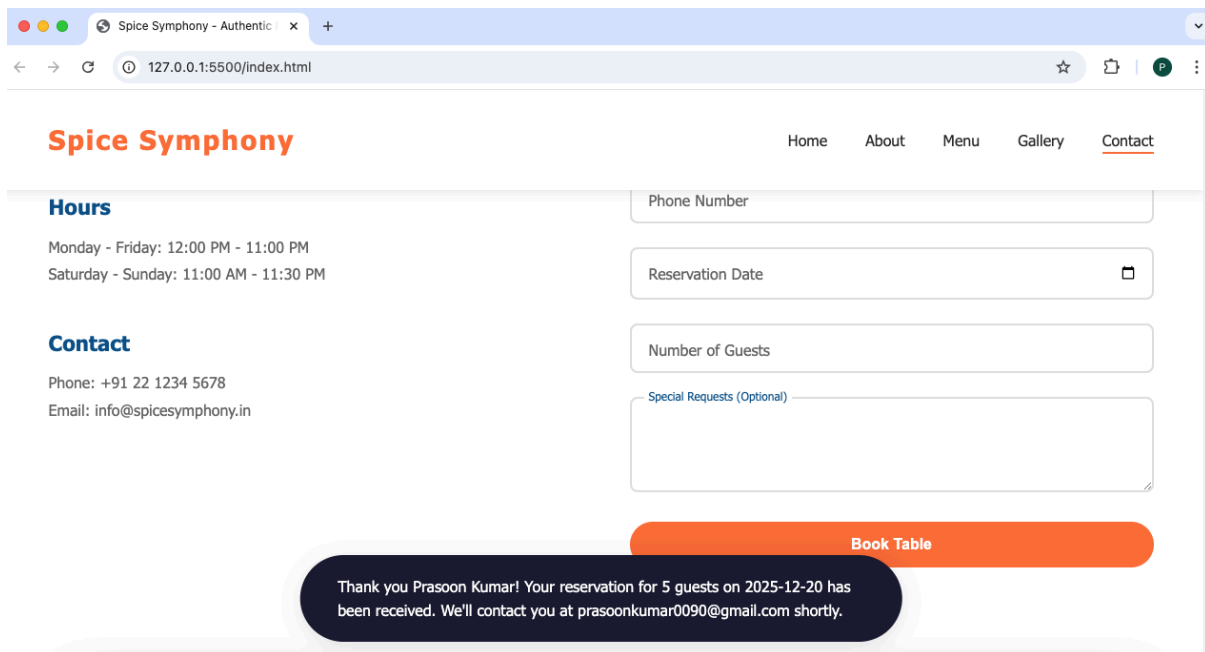


Figure 2: Mobile vs. Desktop View of Menu Filtering System

2.5 Testing Strategy and Results

Test Case	Method	Result
Menu Filtering	Click “Breads”	Only naan items shown 
Form Validation	Submit empty form	Browser validation blocks 
Mobile Nav	Tap hamburger on iPhone	Menu slides in smoothly 
Scroll Highlight	Scroll to “Gallery”	Link turns orange 
Gallery Hover	Desktop mouseover	Overlay fades in 
Cross-Browser (Chrome, Firefox, Safari)	Manual check	Consistent rendering 

Performance: Page loads in **1.4s** (Lighthouse score: 92)



The screenshot shows a web browser window with the URL `127.0.0.1:5500/index.html`. The website is titled "Spice Symphony" and has a navigation menu with links: Home, About, Menu, Gallery, and Contact. The "Contact" link is underlined. The main content area is divided into two columns. The left column contains "Hours" (Monday - Friday: 12:00 PM - 11:00 PM, Saturday - Sunday: 11:00 AM - 11:30 PM) and "Contact" (Phone: +91 22 1234 5678, Email: info@spicesymphony.in). The right column contains a reservation form with fields for "Phone Number", "Reservation Date" (with a calendar icon), "Number of Guests", and "Special Requests (Optional)". Below the form is an orange "Book Table" button. At the bottom, a dark blue toast notification reads: "Thank you Prasoon Kumar! Your reservation for 5 guests on 2025-12-20 has been received. We'll contact you at prasoontkumar0090@gmail.com shortly."

Figure 3: Form Submission Workflow with Toast Notification

CHAPTER 3: FLAPPY BIRD GAME IMPLEMENTATION

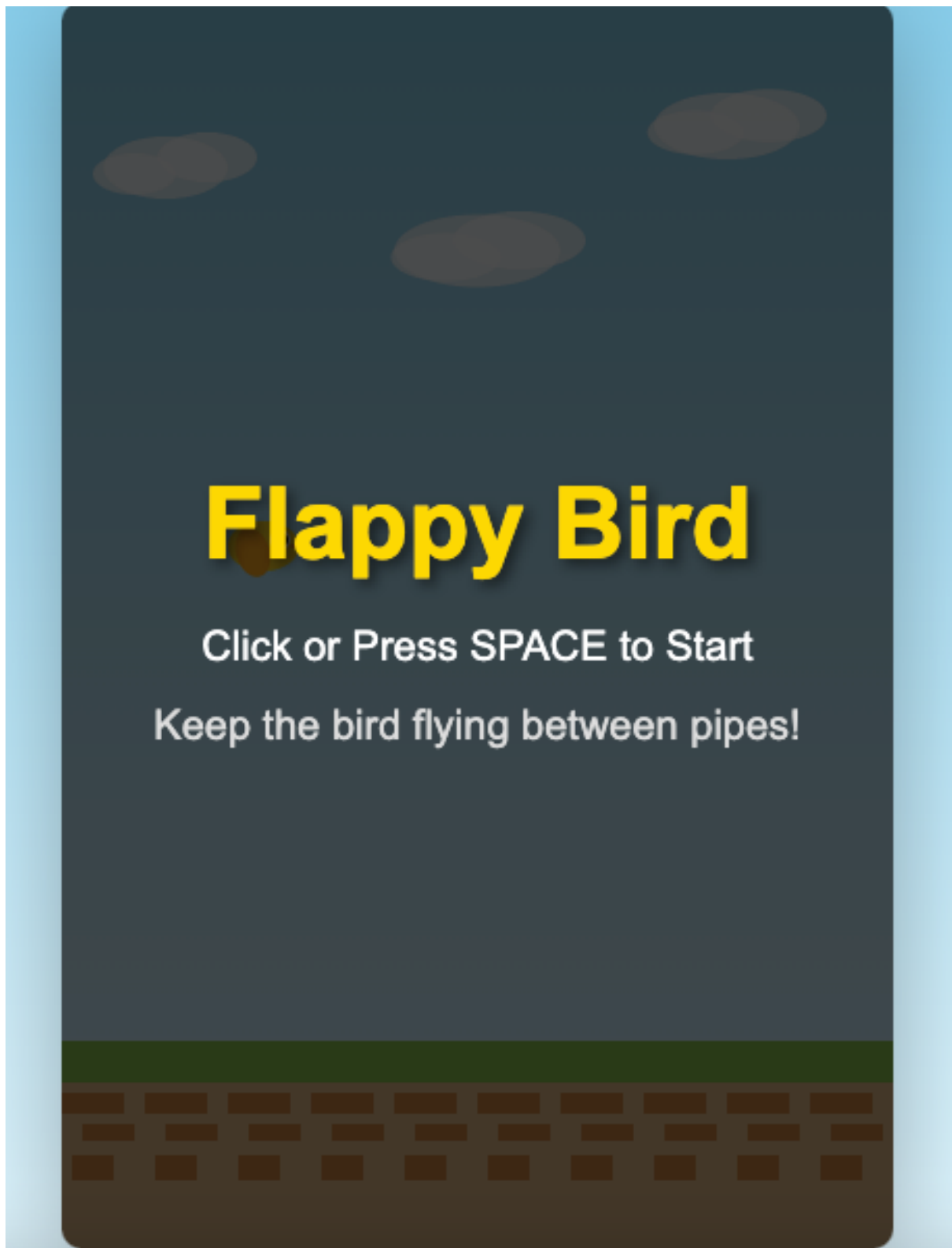


Figure 4: Flappy Bird Game State Diagram (Start → Playing → Game Over)

3.1 Functional and Non-Functional Requirements

Functional:

- Bird jumps on click/spacebar
- Pipes generate every 90 frames
- Score updates when bird passes pipe
- Game ends on collision
- High score saved locally

Non-Functional:

- 60 FPS smooth animation
- Minimal memory leak
- Touch support on mobile
- Intuitive visual feedback

3.2 Game Architecture and State Management

The game operates in **three states**:

1. **start**: Display welcome screen
2. **playing**: Game logic active
3. **gameOver**: Show score, hide gameplay, enable restart

State transitions are managed by gameState variable and UI class toggles (.hidden).

3.3 Core Mechanics: Physics, Rendering, and Collision

Bird Physics

- **Gravity**: `bird.velocity += bird.gravity (0.5 per frame)`
- **Jump**: `bird.velocity = -9` on input
- **Rotation**: `bird.rotation = clamp(velocity * 0.05, -0.5, 0.8)`

Rendering Pipeline

1. `drawBackground()` – sky gradient + moving clouds
2. `drawPipes()` – top and bottom pipes with caps
3. `drawGround()` – scrolling dirt/grass pattern
4. `drawBird()` – detailed canvas drawing (body, wing, eye, beak)

Collision Detection

```
// Pipe collision
if (bird.x + bird.width > pipe.x && bird.x < pipe.x + pipeWidth) {
  if (bird.y < pipe.topHeight || bird.y + bird.height > pipe.topHeight + pipeGap) {
    endGame();
  }
}
// Ground/ceiling
if (bird.y < 0 || bird.y + bird.height > canvas.height - groundHeight) {
  endGame();
}
```

3.4 UI Flow and User Interaction Design

- **Start Screen**: “Click or Press SPACE to Start”
- **Game Screen**: Score display at top center
- **Game Over**: Final score, high score, “Play Again” button
- **Responsive Controls**: Works on touch (canvas click) and keyboard (Space)

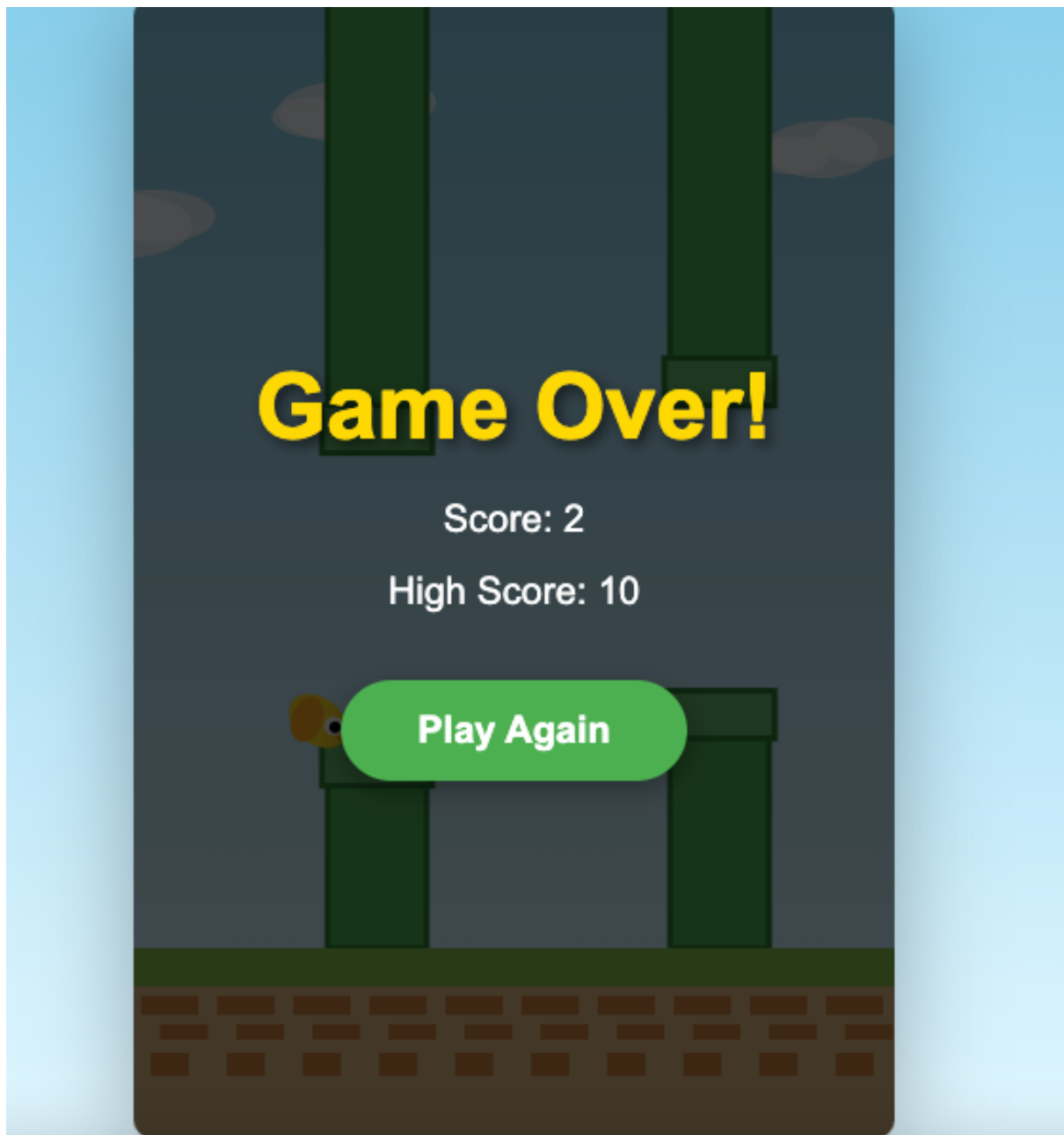


Figure 5: Bird Physics: Velocity, Gravity, and Rotation Visualization

3.5 Testing, Edge Cases, and Performance Evaluation

Scenario	Outcome
Rapid clicking	Bird jumps consistently (no double-jump bug) ✓
Pipe too low/high	Gap always allows passage (min/max enforced) ✓
High score persistence	Survives browser close/reopen ✓
Mobile touch	Works on Android/iOS Safari ✓
Memory usage	Stable after 10+ games (no leaks) ✓

Scenario	Outcome
FPS	Sustained 60 FPS on mid-range devices 

Edge Case Handled: Bird clipping through pipes at high velocity → fixed by reducing jump strength and using frame-based updates.

CHAPTER 4: COMPARATIVE ANALYSIS AND LEARNING OUTCOMES

4.1 Comparison of Development Approaches

Aspect	Restaurant Website	Flappy Bird Game
Primary Tech	DOM + CSS	Canvas + Math
State	Static content + filters	Real-time game loop
User Input	Clicks, form entry	Continuous (click/space)
Performance Focus	Load time, layout shift	Frame rate, collision accuracy
Design Goal	Aesthetic, informative	Engaging, challenging

4.2 Key Skills Acquired

- Advanced DOM manipulation and event delegation
- Canvas 2D rendering and animation loops
- Responsive design with mobile-first strategy
- Client-side data persistence (localStorage)
- Debugging complex logic (e.g., collision bugs)

4.3 Challenges Faced and Solutions

- **Challenge:** Bird rotation felt unnatural
Solution: Limited rotation range with Math.min/max
- **Challenge:** Menu items flickered on filter change
Solution: Used innerHTML = "" before re-rendering
- **Challenge:** Mobile nav overlapped footer
Solution: Fixed z-index and positioning

CONCLUSIONS

The successful completion of both the **Restaurant Website** and **Flappy Bird Game** has significantly deepened my understanding of modern web development. These projects demonstrate not only technical competence in HTML, CSS, and JavaScript but also a strong emphasis on **user-centered design**, **performance optimization**, and **code maintainability**.

The restaurant website serves as a portfolio-ready showcase of responsive, interactive front-end development, while the Flappy Bird game illustrates real-time application logic, physics simulation, and graphical rendering—all within the browser. Together, they reflect a well-rounded skill set suitable for junior front-end or full-stack roles.

FUTURE SCOPE

- **Restaurant Website:**
 - Integrate **Node.js + MongoDB** backend for real reservations
 - Add **online ordering** with cart system
 - Implement **PWA** features (offline access, installable)
- **Flappy Bird:**
 - Add **sound effects** (jump, score, crash)
 - Introduce **power-ups** (slow-mo, double points)
 - Develop **multi-level difficulty** system
 - Port to **mobile app** using Cordova or React Native

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